Supporting Information for General Framework for Geometric Deep Learning on Tensorial Properties of Molecules and Crystals

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$E(3)$ group equivariance and equivariant graph neural networks

Atomistic systems can be represented as coordinate systems in three-dimensional (3D) Euclidean space. Transformations that preserve Euclidean distance between any two points in 3D space form the $E(3)$ group, which includes translation $\hat{T}$, rotation $\hat{R}$ and inversion $\hat{I}$. Furthermore, $\hat{T}$ and $\hat{R}$ form the $SE(3)$ group, while the $SO(3)$ group only considers rotation symmetry and the $O(3)$ group consists of $\hat{R}$ and $\hat{I}$; these are subgroups of $E(3)$. Properties of atomistic systems are $O(3)/SO(3)$-equivariant to geometric transformations, because laws of physics are independent of the choice of coordinate systems. Therefore, machine learning methods for 3D atomistic systems should be $E(3)/SE(3)$-equivariant to geometric transformations acting on atom positions $\{\vec{r}_A\}$, in order to respect the fundamental symmetry.

The conventional strategy of using relative position vectors between atoms A and B, $\{\vec{r}_{AB}\}$, in graph representation of molecules and crystals ensures the translational equivariance for GNN models. Therefore, equivariant GNN models differ from invariant models primarily by further satisfying rotational and inversional equivariance. That is why some equivariant GNN models in the literature claim to be $O(3)$ or $SO(3)$-equivariant with no emphasis on the translational equivariance.

Equivariant GNN models using the spherical basis can represent hidden features decomposed and transformed as the direct sum of the irreducible representations (Irreps) of the $O(3)/SO(3)$ group (**Eq**. (1) in the main text). The initial features are generated by the spherical harmonics (SPHs) $Y^{(l)}(\hat{r})$, where $\hat{r}$ is the unit vector $\hat{r} \equiv \vec{r}/\|\vec{r}\|$. The SPHs form a basis for the Irreps of the $SO(3)$ group. When used to project $\{\vec{r}_{AB}\}$, these features are inherently $SO(3)$-equivariant, and can be used to construct $SE(3)$-equivariant features.[1] Meanwhile, SPHs are also irreducible representations of the $O(3)$ group, where the parity of $Y^{(l)}(\hat{r})$ is given by $(-1)^l$.[2] Therefore, the spherical features expanded by SPHs are exactly

$O(3)$-equivariant, and can be $E(3)$-equivariant, if the parity for each l channel is explicitly accounted for in the feature construction and if the model processes relative coordinates for translational invariance. In other words, while commonly termed $SE(3)$-equivariant, these models[3–5] are in fact $E(3)$-equivariant, with equivariant features consisting solely of polar tensor components, due to the intrinsic physical nature of SPHs.

To construct the initial set of node equivariant features of atom A, $\boldsymbol{\chi}_A^0$, in an equivariant GNN, a message passing operation involving aggregation over its neighboring nodes $\mathcal{N}(A)$ is commonly employed:[4]

$$\boldsymbol{\chi}_A^0 = \sum_{B \in \mathcal{N}(A)} \phi_m^0 \left(\boldsymbol{x}_A^0, \boldsymbol{x}_B^0, \vec{r}_{AB} \right) \tag{S1}$$

Here, ϕ_m^0 is the message function in the first message passing layer of the GNN, where the node feature of A, $\boldsymbol{x}_A^0$, is nothing but invariant scalar embeddings ($l = 0$). For subsequent layers ($t > 0$), where both equivariant and invariant node features, $\boldsymbol{\chi}^{t-1}$ and $\boldsymbol{x}^{t-1}$, are propagated jointly, the updated equivariant node features $\boldsymbol{\chi}_A^t$ are computed as:

$$\boldsymbol{\chi}_A^t = \sum_{B \in \mathcal{N}(A)} \phi_m^t \left(\boldsymbol{x}_A^{t-1}, \boldsymbol{x}_B^{t-1}, \boldsymbol{\chi}_A^{t-1}, \boldsymbol{\chi}_B^{t-1}, \vec{r}_{AB} \right) \tag{S2}$$

To construct full $E(3)$-equivariant representations of the graph, tensor products respecting the $O(3)$ group equivariance are further required to include parity-wise channels in the message function ϕ_m^t of the deeper layers, resulting in what are commonly referred to as $E(3)$-equivariant models in the literature:[6–9]

$$\begin{aligned} \omega_{m_3}^{(l_3,p_3)} &= \left(\boldsymbol{\kappa}^{(l_1,p_1)} \otimes \boldsymbol{\tau}^{(l_2,p_2)} \right)_{m_3} \\ &= \sum_{m_1=-l_1}^{l_1} \sum_{m_2=-l_2}^{l_2} C_{(l_1,m_1)(l_2,m_2)}^{(l_3,m_3)} \kappa_{m_1}^{(l_1,p_1)} \tau_{m_2}^{(l_2,p_2)} \end{aligned} \tag{S3}$$

Here, $C_{(l_1,m_1)(l_2,m_2)}^{(l_3,m_3)}$ are the Clebsch-Gordan (CG) coefficients, and the parity is determined according to $p_3 = p_1 \cdot p_2, p \in \{-1, +1\}$.

On the other hand, the XPaiNN architecture[10] is $SE(3)$-equivariant under the commonly

accepted definitions outlined above. Its equivariant features are expanded using SPHs, and no CG tensor product operations are employed to generate parity-resolved channels for l-order Irrep features. Recently, this design philosophy that follows PaiNN[3] is gaining renewed attention in developing efficient and effective $SE(3)$-equivariant GNN models.[11,12] Despite certain controversy, here and in the following text, we adopt the convention: models constructing equivariant features consisting solely of polar tensor components, such as XPaiNN, are classified as $SE(3)$-equivariant, whereas models such as NequIP[6] and its successors are considered $E(3)$-equivariant, as they incorporate multiple $O(3)$-equivariant tensor product operations within their message passing layers and construct geometric features with full $E(3)$-group equivariance, i.e., those hold parity-wise representations of both polar and axial (pseudo) tensor components.

Design of Cartesian tensor output via customized tensor product and its validation

The challenge in machine learning tensorial properties lies in satisfying both the equivariance under coordinate transformations and the intrinsic symmetry of the physical tensors themselves. Equivariant models offer a significant advantage in learning and predicting tensorial targets due to their inherent ability to correctly model equivariant objects. To date, several strategies have been proposed to construct Cartesian tensor outputs from the feature representations generated by equivariant GNNs.

As pointed out in the main text, PaiNN[3] and ETGNN[13] proposed to construct order-N tensors by performing multiple tensor products of ($l = 1$) vectors generated by their message passing blocks. While this strategy ensures the equivariance of tensor targets, it lacks the generality to satisfy the fundamental symmetry requirement in more complex scenarios, such as the order-3 hyperpolarizability and the order-4 elastic tensor.

The work of FIREANN[14] calculated tensorial properties based on their physical derivative definitions as responses to external fields or perturbations. For instance, the dipole moment $\boldsymbol{\mu}$ and polarizability $\boldsymbol{\alpha}$ are modeled as the first and second order derivatives of energy E with respect to electric field $\vec{\epsilon}$, respectively. Here E is the network output and $\vec{\epsilon}$ is an artificial atomic field vector provided as conditional input.[14] In other words, the tensorial properties themselves are not directly the network outputs. The key advantage of such a strategy is that the exact physical constraints of the target property are guaranteed by the differential operator (for equivariance) and the conditional input of differential variables (for fundamental symmetry), as pointed out by Kozinsky and coworkers.[15] More recently, GMTNet[16] introduced a general strategy that is fully consistent with the intrinsic crystal symmetries, expressed as

$$\mathbf{T} = \frac{\partial \left(\mathbf{G} \otimes \boldsymbol{\epsilon} \right)}{\partial \boldsymbol{\epsilon}} \tag{S4}$$

where $\mathbf{G}$ is the crystal-level equivariant feature output by the neural network via average

node aggregation, and $\boldsymbol{\epsilon}$ is a target-wise conditional input. This input acts as the electric field vector when modeling the dielectric tensor, and as the 3×3 strain tensor when modeling the piezoelectric tensor and the elastic tensor, respectively. GMTNet demonstrated that once the crystal-level feature $\mathbf{G}$ satisfies the symmetry of the corresponding crystal space group, then the intrinsic symmetry of the target tensor property can be fully preserved via **Eq**. (S4).[16] Meanwhile, the disadvantage for the differential strategy lies in its computational inefficiency in both training and inference. Calculating gradients of the parameters with respect to the conditional input can be resource-intensive, especially for atom-wise tensor properties, a drawback that has not been clearly addressed in these works.

Noticing the mathematical fact that the permutation symmetry of Cartesian tensors can be mapped to a unique direct sum of Irreps, we adopt, in the current work, the strategy of directly modeling the spherical representation of tensor targets and then transforming it into the Cartesian form. This idea is conceptually consistent with those used in DetaNet,[17] MatTen[9] and the related works.[18,19] In these models, the targeted spherical representations are constructed either via equivariant linear transformation ($O(3)$-Linear)[9,19] or by spherical harmonic expansion of invariant hidden features.[17,18] Building upon the original PaiNN model,[3] we previously designed a uniform framework for constructing tensorial outputs for the XPaiNN model,[10] which employed an idea of the gated equivariant block[3] in order to take joint advantage of the equivariance and nonlinearity from the equivariant and invariant features, respectively, in the XPaiNN architecture, as depicted in **Fig**.S1 (b). In practice, this "gated" output module has proven effective in learning and predicting various tensorial properties that are polar tensors by definition. However, we encountered limitations when applying it to tensors with explicit pseudo components, especially in crystalline systems, as the XPaiNN architecture lacks pseudo-tensor features in its equivariant representations, as discussed above in Section S1. Moreover, this design lacks the flexibility in modeling higher-order tensors, as the hidden feature representations are tightly constrained by the target tensor. This leads to a rapid increase in model parameters with the tensor order, a

known drawback also observed in $E(3)$ models adopting the linear transformation layers ("lin") for output generation.

To address these issues, we design and introduce in this work the **node-wise self-mix layer**, a novel and general-purpose module for constructing tensorial properties of arbitrary order and permutation symmetry. This design leverages the expressive power of tensor product operation to generate equivariant outputs, extending the idea previously suggested in PaiNN,[3] where repeated tensor products of vector features are proposed to build higher-order tensors. As discussed in the main text, our tensor product layer decouples the representation constraints between the hidden equivariant features and the target output, overcoming the limitations of both the "gated" and "lin" designs and enabling $SE(3)$ models to learn and predict tensors containing pseudo components. Furthermore, by customizing the tensor product operation, our "mix" layer is broadly applicable across diverse systems, including atoms, molecules, and periodic crystals. It provides a flexible, efficient, and direct solution for modeling various tensorial properties of arbitrary order and fundamental symmetry, as demonstrated in the following sections.

In constructing the **node-wise self-mix layer**, we first use a customized tensor product operation[2] to construct the equivariant node feature $\boldsymbol{\chi}_A^o = \bigoplus \boldsymbol{\omega}_A^{(l_o,p_o)}$ as a direct sum of the desired Irreps from the equivariant hidden feature $\boldsymbol{\chi}_A^f$. This operation, referred to as the self-mix of node equivariant features, is defined as:

$$\boldsymbol{\omega}_{A,u}^{(l_o,p_o)} = \sum_{(l_1,p_1)} \sum_{(l_2,p_2)} \boldsymbol{\omega}_{A,u}^{f,(l_1,p_1)} \otimes \boldsymbol{\omega}_{A,u}^{f,(l_2,p_2)} \tag{S5}$$

For simplicity and efficiency, we restrict the tensor product operation to act within the same feature channel index u of $\boldsymbol{\omega}$ (i.e., channel-wise). As a result, the Irreps layout of the layer output $\boldsymbol{\chi}_A^o$ (e.g. 32×0e + 32×1e + 32×2e) may differ from that of the target Cartesian tensor (e.g. 1×0e + 1×1e + 1×2e, corresponding to an asymmetric order-2 tensor). Consequently, linear transformations are required within each $\boldsymbol{\omega}^{(l_o,p_o)}$. Such transformations

can be equivalently integrated into the tensor product operation defined in **Eq**. (S5), acting on each admissible path of tensor products that yields $(l_1, p_1) \otimes (l_2, p_2) \rightarrow (l_o, p_o)$ with additional parameters w as the path-wise weights for linear transformation, enabling dimensionality adjustment and feature mixing within each equivariant channel:

$$\boldsymbol{t}_{A,w}^{(l_o,p_o)} = \sum_{(l_1,p_1)} \sum_{(l_2,p_2)} \sum_{u} w_{uuw}^{(l_o,p_o;l_1,p_1;l_2,p_2)} \circ \boldsymbol{\omega}_{A,u}^{f,(l_1,p_1)} \otimes \boldsymbol{\omega}_{A,u}^{f,(l_2,p_2)} \tag{S6}$$

In the current design, however, we do not introduce these additional parameters, w. Instead, we utilize the transformations of the invariant node feature $\boldsymbol{x}_A^f$ of XPaiNN to serve as the path-wise weights, as described in the main text. Thus, we arrive at the final working formula of the **node-wise self-mix layer** as

$$\boldsymbol{t}_{A,w}^{(l_o,p_o)} = \sum_{(l_1,p_1)} \sum_{(l_2,p_2)} \sum_{u} \mathrm{MLP}\left(\boldsymbol{x}_A^f\right)_{uuw}^{(l_o,p_o;l_1,p_1;l_2,p_2)} \circ \boldsymbol{\omega}_{A,u}^{f,(l_1,p_1)} \otimes \boldsymbol{\omega}_{A,u}^{f,(l_2,p_2)} \tag{S7}$$

In Eq. (S7), $\mathrm{MLP}\left(\boldsymbol{x}_A^f\right)_{uuw}^{(l_o,p_o;l_1,p_1;l_2,p_2)}$ denotes the output of a multi-layer perceptron (MLP) acting on the invariant feature $\boldsymbol{x}_A^f$, which is subsequently reshaped and partitioned into path-specific scalars corresponding to each admissible combination of $(l_o, p_o; l_1, p_1; l_2, p_2)$ and the indices uuw. The output dimension of the MLP is set to the total number of valid paths counted for the tensor product operation. This formulation aligns with **Eq**. (2) and **Fig**. 1(d) in the main text, and avoids explicit linear projection layers by using a context-aware MLP to generate path-specific mixing coefficients, thereby achieving implicit dimensionality reduction conditioned on local atomic environments. Thus, built upon the aforementioned strategies (i.e., **Eq**. (S5) to (S7)), this self-mix layer for feature transformation provides a more general and expressive approach for constructing arbitrary tensor targets at both local and global levels via GNN models.

At this point, it is quite straightforward to prove that our strategy naturally satisfies both the equivariance and the fundamental symmetry requirements for Cartesian tensor output, by

noticing the fact that the equivariant feature generated by an equivariant GNN transforms under geometric operations, such as rotation $\hat{R}$, as follows:

$$\boldsymbol{\chi}' = \bigoplus_l \mathbf{W}_D^l \left(\hat{R}\right) \circ \boldsymbol{\omega}^l = \hat{R}\, \boldsymbol{\chi} \tag{S8}$$

Here, $\mathbf{W}_D^l \left(\hat{R}\right)$ refers to the l-order Wigner-D matrix representation of rotation $\hat{R}$ and $\circ$ denotes matrix multiplication. Consequently, the Cartesian tensor output of an order-N tensor under the Cartesian transformation $\mathbf{Q}$ (whose last dimension is in consistency with the Irreps layout of the tensor) is given by

$$\begin{aligned} \mathbf{T}' &= \mathbf{Q} \circ \sum_{A \in \mathcal{G}} \boldsymbol{T}'_A \\ &= \mathbf{Q} \bigoplus_l \mathbf{W}_D^l \left(\hat{R}\right) \mathbf{Q}^{\mathrm{T}} \circ \sum_{A \in \mathcal{G}} \mathbf{Q} \boldsymbol{t}_A^l \\ &= R_{1i} R_{2j} \ldots R_{Nr} \mathbf{T}_{ij\ldots r} \end{aligned} \tag{S9}$$

Note that $\mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathbf{I}$ and $\mathbf{Q}\mathbf{W}\mathbf{Q}^{\mathrm{T}}$ transforms the Wigner-D matrix to its Cartesian representation. Furthermore, since $\mathbf{T}$ is modeled in our strategy to directly satisfy the required Irreps, it inherently fulfills the fundamental symmetry requirement (i.e., the permutation symmetry of indices) of the target property.

For the more challenging case of satisfying the intrinsic symmetry of tensors, which originates from the system symmetry (primarily in crystals), a proof similar to that in GMTNet[16] can be established. Specifically, if the representation of the crystal graph fully reflects the symmetry of the space group to which the crystal belongs, then our strategy inherently respects the intrinsic symmetry requirement of the target tensor. This is succinctly

expressed as

$$\begin{aligned}\mathbf{T}' &= \mathbf{Q} \circ \sum_{A \in \mathcal{G}} \boldsymbol{T}'_A \\ &= \mathbf{Q} \circ \bigoplus_{l} \mathbf{W}_D^l \left(\hat{R}_i \right) \boldsymbol{T}_{\mathcal{G}} \\ &= \mathbf{T}\end{aligned} \tag{S10}$$

Here, $\hat{R}_i$ represents a symmetry operation that belongs to the space group and $\boldsymbol{T}_{\mathcal{G}}$ denotes the graph-level equivariant feature. Under the condition that the GNN model employed reflects the full symmetry of the crystal space group, the last equivalence relationship in **Eq**. (S10) can naturally be established. This can be readily understood and formally established for $E(3)$-equivariant models.[16] For the $SE(3)$ graph representations of polar tensors, we shall point out here that the same results still hold, as long as the model correctly captures the true inversion behavior of the system. This requires the inclusion of odd-l Irreps (e.g., $l = 1, 3, 5, \ldots$), being non-trivial under inversion transformation. In other words, the output module designed in this work can generate a target tensor of arbitrary order and intrinsic symmetry, while ensuring full $O(3)$-equivariance. This capability is achieved when the output module is combined with either an $E(3)$ or $SE(3)$ GNN model, whose message-passing and update blocks can faithfully reflect the full symmetry of the system.

Table S1: Numerical validation of equivariance for XPaiNN combined with the current output module in prediction of Cartesian tensors. The analysis is performed in terms of $L2$-norm distance.

Property	Origin	Rotate 45°-x	Rotate 45°-y	Rotate 45°-z
$\boldsymbol{\mu}$	0.0118	0.0118	0.0118	0.0118
$\boldsymbol{\alpha}$	0.2071	0.2071	0.2071	0.2071
$\boldsymbol{\beta}$	10.15	10.15	10.15	10.15
$\boldsymbol{C}$	48.71	48.71	48.71	48.71

To further substantiate the proof of **Eq**. (S9), we examined whether the numerical results for Cartesian tensors predicted by XPaiNN satisfy equivariance under rotations. To this end, we generated rotated data from the test sets of QM9S and MatTen datasets used in the

current work. Specifically, both the structures of molecules or crystals and their corresponding tensorial properties were rotated by 45° around the x, y and z axes, respectively. We evaluated the performance by calculating the $E(3)$-invariant $L2$-norm distance between the predictions and the ground truth. This analysis was applied comprehensively to tensorial properties ranging from order 1 to 4. The models examined here were not retrained, but were directly taken from those reported in the main text, i.e., they were trained only on the original datasets.

As summarized in **Table** S1, our strategy of constructing Cartesian tensors fully satisfies the equivariance requirements in prediction of Cartesian tensors in all cases. This indicates that the proof of **Eq**. (S9) is valid for general cases of order-N Cartesian tensors.

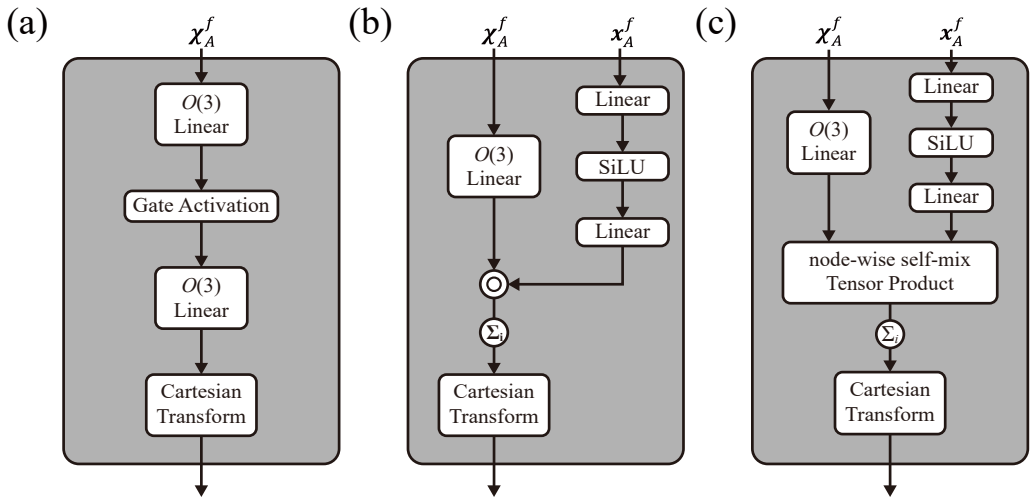


Figure S1: **Output modules implemented for predicting Cartesian tensor properties.** **(a)** The "lin" module representing the most common approach in the literature, the gate activation block is introduced to enhance model capacity. **(b)** The "gated" module designed in the orignal XPaiNN work[10] for predicting Cartesian tensor targets. **(c)** The "mix" module introduced in this work, with the **node-wise self-mix tensor product** responsible for constructing tensorial outputs of arbitrary order and fundamental symmetry.

In order to compare the performances of different direct approaches for constructing Cartesian tensor outputs, here we evaluate the results of three output modules, namely "lin" (the most common one), "gated" (the original scheme used in XPaiNN) and "mix" (the current design), combined with either XPaiNN or the Test Model introduced in the main

text. As illustrated in **Fig**. S1, we replace the common single linear transformation scheme with an equivariant gate activation block[4] to enhance model capacity. The "gated" output module can be viewed as a joint and enhanced version of both the linear scheme and the spherical harmonic expansion approach (via gated Hadamard product between invariant and equivariant features[10]). All models are trained with the same hyperparameter settings including the hidden feature representations, thus the performances can be fairly compared between different approaches. Given the limitations of the "gated" scheme in handling pseudo-tensor components when used with XPaiNN, here we focus on properties that can be fully described by polar tensor features, i.e., properties in QM9S and elastic tensors of the MatTen dataset.

Table S2: Performances of various output schemes combined with XPaiNN or the Test Model in predicting tensorial properties in the QM9S dataset.

Property[a]	Metrics [Unit][b]	$SE(3)$-Test-lin	$SE(3)$-Test-mix[c]	XPaiNN-gated	XPaiNN-mix
$\boldsymbol{\mu}$	MAE [Debye]	0.0162	0.0116	0.0058	**0.0051**
	RMSE [Debye]	0.0238	0.0172	0.0099	**0.0085**
$\boldsymbol{\alpha}$	MAE [a_0^3]	0.1698	0.1410	0.0556	**0.0530**
	RMSE [a_0^3]	0.3006	0.2506	0.0993	**0.0949**
$\boldsymbol{\beta}$	MAE [a.u.]	3.0715	2.2128	1.4254	**1.2561**
	RMSE [a.u.]	6.9738	4.7501	2.7242	**2.4417**
$\boldsymbol{D}$	MAE [D $\cdot$ Å]	0.6209	0.5165	0.2647	**0.2213**
	RMSE [D $\cdot$ Å]	1.2132	1.0423	0.6711	**0.5977**
$\boldsymbol{O}$	MAE [D $\cdot$ Å^2]	10.1943	8.9775	4.3939	**4.1191**
	RMSE [D $\cdot$ Å^2]	24.9544	22.4674	15.3516	**15.2100**

[a] Dipole moment $\boldsymbol{\mu}$, polarizability $\boldsymbol{\alpha}$, first hyperpolarizability $\boldsymbol{\beta}$, quadruple moment $\boldsymbol{D}$ and octupole moment $\boldsymbol{O}$.
[b] Best results are in **bold**.
[c] The MAE results are reported in **Table** 1 of the main text.

Table S2 and **Table** S3 summarize the performances of different output schemes combined with either XPaiNN or the Test Model in predicting tensorial properties in the QM9S and the MatTen datasets, respectively. Clearly, our current approach, namely the XPaiNN-mix model, shows leading performances in all examined cases. Meanwhile, XPaiNN-gated is also competitive in predicting elastic properties, suggesting the capacity of the XPaiNN

Table S3: Performances of various output schemes combined with XPaiNN or the Test Model in predicting elastic tensor and elastic properties in the MatTen dataset.

Property[a]	Metrics [Unit][b]	$SE(3)$-Test-lin[c]	$SE(3)$-Test-mix[c]	XPaiNN-gated	XPaiNN-mix[c]
$\boldsymbol{C}$	Fnorm ↓ [GPa]	41.83	43.95	33.83	**31.51**
	EwT 25% ↑ [%]	80.9	81.3	88.4	**89.8**
	EwT 10% ↑ [%]	42.9	42.0	57.0	**62.2**
	EwT 5% ↑ [%]	16.8	14.4	30.1	**31.5**
K	MAE [GPa]	8.41	8.57	5.64	**5.40**
	RMSE [GPa]	15.82	14.22	11.41	**10.45**
G	MAE [GPa]	11.05	7.86	6.38	**6.33**
	RMSE [GPa]	69.58	12.67	**11.10**	11.23
E	MAE [GPa]	22.84	18.50	14.86	**14.74**
	RMSE [GPa]	71.42	29.78	**26.14**	26.36

[a] Elastic tensor $\boldsymbol{C}$ (in the 6×6 stiffness matrix form), bulk modulus K, shear modulus G and Young's modulus E.
[b] Frobenius norm distance (Fnorm) and High-quality prediction rate (EwT) follow the definition in Ref. 16. Best results are in **bold**.
[c] Results have been reported in **Table** 5 of the main text.

architecture. To our notice here, as also discussed in the main text, the $SE(3)$-Test-mix model shows superior precision over its "lin" counterpart in molecular tensorial properties and more robust error distributions for elastic properties, highlighting that the success of the XPaiNN-mix model is a joint triumph of both the model architecture and the current **node-wise self-mix** approach.

Further validation of the method and comparisons between different approaches, including fulfillment of intrinsic symmetry in crystalline systems, predicting derived properties of molecules, as well as training efficiency, are provided in the following sections.

Symmetry issues in learning crystal properties

When training models on single systems with perturbed structures, which are study cases previously considered in the ETGNN model,[13] we observed that XPaiNN, combined with the "mix" module, produced nonphysical predictions for properties like piezoelectric tensor $\boldsymbol{e}$ and Born effective charge $\boldsymbol{Z}^*$, if the hidden equivariant features are truncated at $l_{\max} = 2$. This behavior is illustrated in the leftmost column in **Fig**. S3. We soon noticed that such a behavior is quite similar to the PhiSNet[5] ($SE(3)$-equivariant) prediction on crystal Hamiltonian matrix, reported by the work of HamGNN.[8] In that work, the authors attribute the failure to the representations generated from the $SE(3)$-equivariant architecture being unable to distinguish the inversion symmetry. As widely recognized in the development of equivariant GNNs, capturing such physical behaviors requires full $E(3)$-equivariance.[16,19]

To address this limitation, we introduce a parity expansion layer within the "mix" output module. This layer performs an additional self-mix tensor product, structured similarly to previous layers but with shared parameters for linear transformation within each tensor product path of $(l_1, p_1) \otimes (l_2, p_2) \to (l_o, p_o)$

$$\boldsymbol{\omega}_{A,u}^{(l_o,p_o)} = \sum_{(l_1,p_1)} \sum_{(l_2,p_2)} a_{uuu}^{(l_o,p_o;l_1,p_1;l_2,p_2)} \circ \boldsymbol{\omega}_{A,u}^{f,(l_1,p_1)} \otimes \boldsymbol{\omega}_{A,u}^{f,(l_2,p_2)} \tag{S11}$$

The tensor product is implemented as described in **Eq**. (S3), and its primary role is to expand parity-wise channels from the network hidden features $\boldsymbol{\chi}^f$ to generate full $E(3)$-equivariant representations, as illustrated schematically in **Fig**.S2 (a) and (b). We refer to this enhanced output module with the "tp" suffix, for the sake of simplicity in subsequent discussions.

However, a direct combination of the "tp" module with the XPaiNN network using hidden features up to $l_{\max} = 2$ yields only marginal improvement in test performances. This is particularly evident in predictions of the piezoelectric tensor of AlN and the BEC at the central atom sites (Al and Si) of Al_2O_3 and SiO_2, as shown in the plots in the third column of **Fig**. S3. Meanwhile, the inclusion of $l_{\max} = 3$ hidden features directly combined with

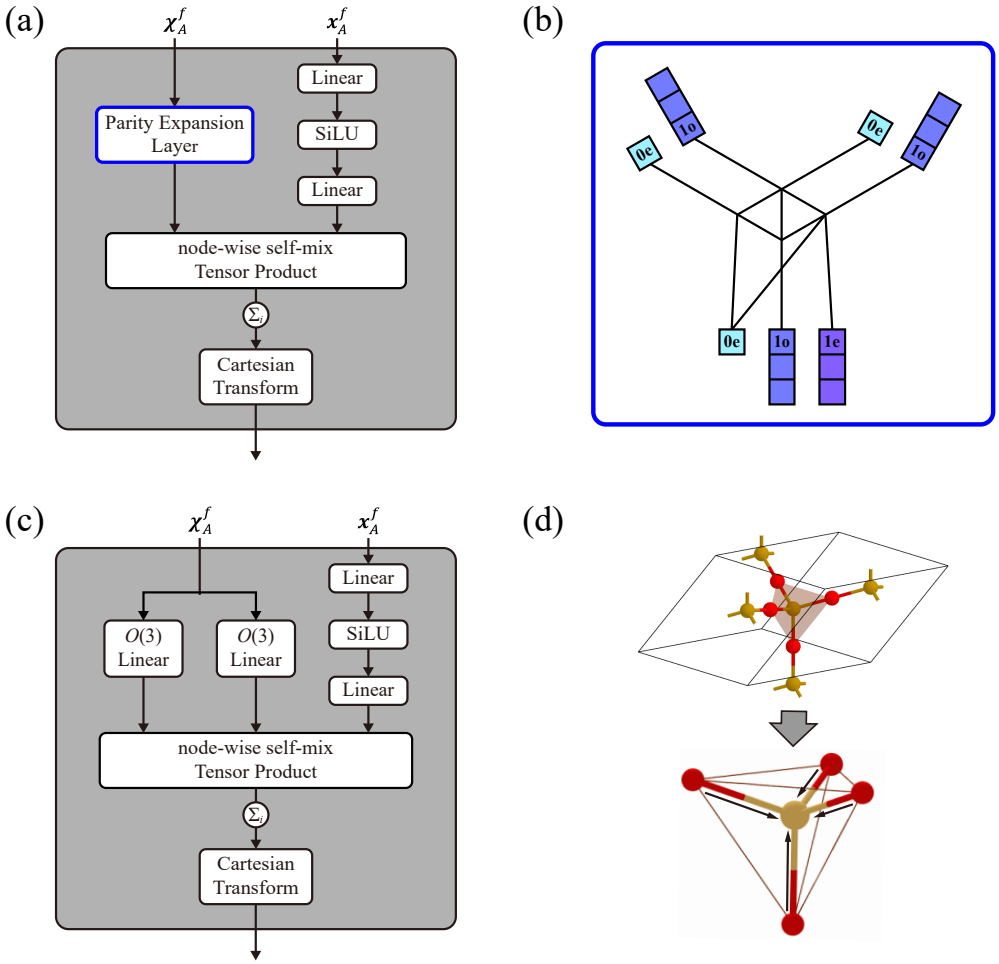


Figure S2: **Output modules designed for parity issues in crystalline systems. (a)** The parity expansion module "tp", the parity expansion layer is introduced to replace the $O(3)$ Linear in the original "mix" output module. **(b)** Illustration of the channel-wise parity expansion tensor product with a truncation at $l_{\max} = 1$. **(c)** The "mix2" module modified from the original "mix", another $O(3)$ Linear layer is invoked to construct non-equivalent entries to the self-mix tensor product. **(d)** Issues of directional message passing in a highly symmetric local chemical environment, the Irreps feature of `1o` channel of the central site remains zero due to symmetry and the sum-up aggregation.

the original "mix" scheme successfully resolves the issues in piezoelectric tensor, but no improvement is observed in the BEC prediction, as seen in the second column of **Fig**. S3. Ultimately, all nonphysical predictions are eliminated only when both modifications are combined: using the "tp" module and $l_{\max} = 3$ hidden feature expansions in XPaiNN, as demonstrated in the fourth column of **Fig**. S3.

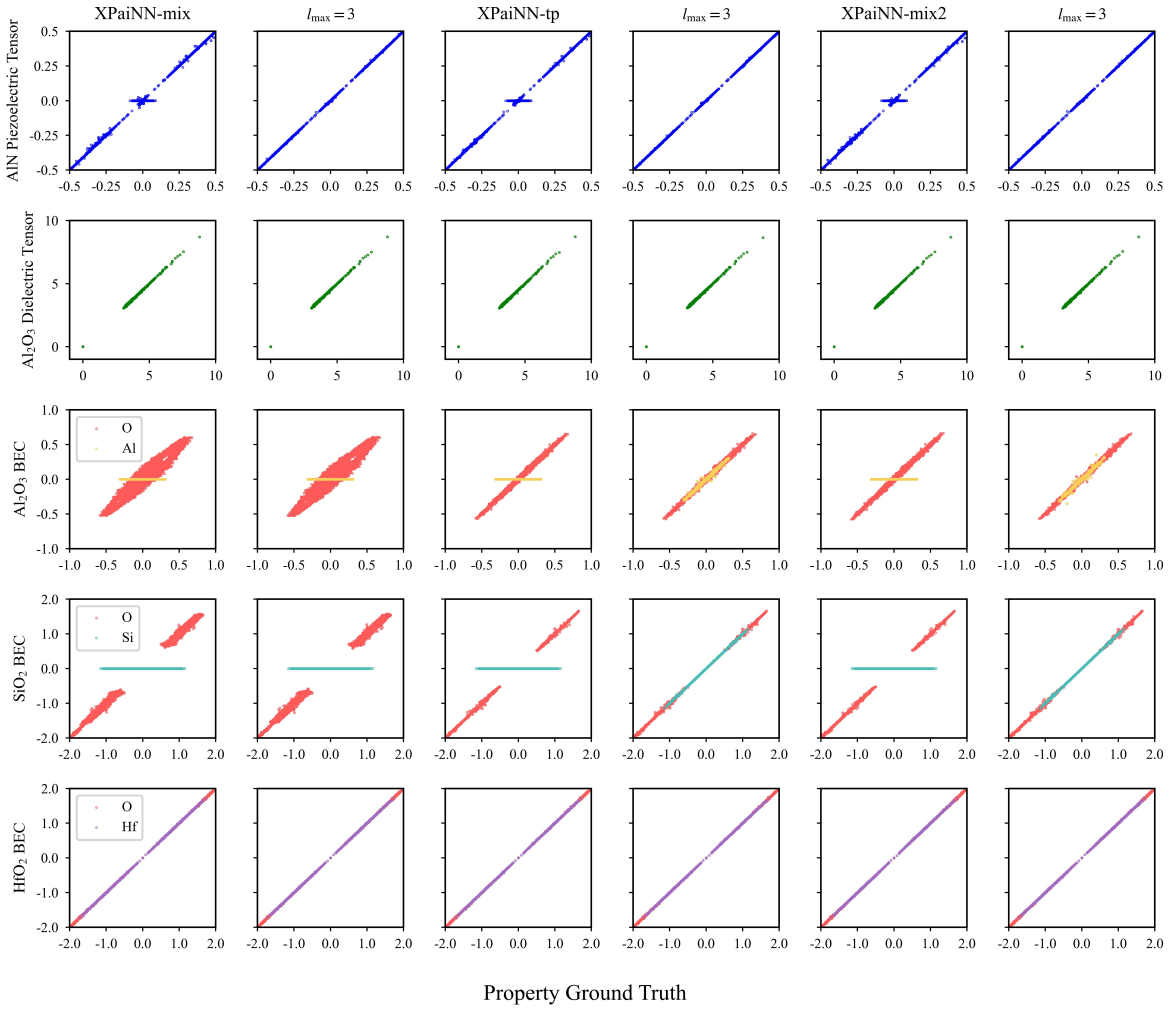


Figure S3: Performances on single systems with perturbed structures, predicted with different hyperparameters. Model differences lie in the maximum l component in hidden Irreps features and the combinations of the output modules among "mix", "tp" and "mix2".

The underlying reason for the observed behavior lies in the interplay between the graph representations and the Irreps of the target properties. The piezoelectric tensor $\boldsymbol{e}$ is an order-3 tensor, whose Irreps consist entirely of odd parity components, i.e., $2\times1o+1\times2o+1\times3o$.

According to the parity selection rule in $O(3)$ tensor products, an odd parity channel can only be generated by tensor products of one odd parity channel and one even parity channel, $\mathsf{o} \otimes \mathsf{e} \rightarrow \mathsf{o}$. In graph representations based on polar tensors with hidden features truncated at $l_{\max} = 2$, the only available odd-parity channel is $\mathsf{1o}$. However, in a highly symmetric local chemical environment, such as the Al site at the center of a tetrahedron formed by four surrounding N atoms in AlN, and vice versa, the directional message passing and sum-up aggregation mechanism in GNNs lead to an all-zero representation in the $\mathsf{1o}$ channel of the node equivariant feature, as illustrated in **Fig**.S2 (d). This perfect cancellation in the $\mathsf{1o}$ channel occurs because opposite bond vectors contribute with opposite signs. Consequently, the final graph representation of AlN lacks any odd-parity signal in its $\mathsf{1o}$ channel, leading to the nonphysical prediction of $\boldsymbol{e}$. In contrast, expanding the feature space to include $l = 3$ ($\mathsf{3o}$) introduces a new odd-parity channel that is less susceptible to complete cancellation due to the more complex angular dependence on spherical harmonics. When the model is combined with either the "mix" or the "tp" module, these non-trivial contributions from the odd parity channel can thus be generated via tensor products, e.g., via $\mathsf{3o} \otimes \mathsf{2e}$, thereby restoring physical consistency in the predicted piezoelectric tensor.

The challenges in predicting BEC tensor $\boldsymbol{Z}^*$ are related but involve additional complexity, as $\boldsymbol{Z}^*$ is an atomic local feature and is directly affected by the symmetry and completeness of the local representation of the node. The Irreps of $\boldsymbol{Z}^*$ under $O(3)$ are $\mathsf{1{\times}0e} + \mathsf{1{\times}1e} + \mathsf{1{\times}2e}$, corresponding to an asymmetric order-2 tensor. The antisymmetric part arises from the $\mathsf{1e}$ channel, as is commonly known. Therefore, accurate prediction of $\boldsymbol{Z}^*$ critically depends on the model's ability to capture antisymmetric contributions, which in turn requires a correct and non-vanishing representation in the $\mathsf{1e}$ channel. Within an $O(3)$ tensor framework, the $\mathsf{1e}$ component can be generated by polar tensors from either $\mathsf{1o} \otimes \mathsf{1o}$ or $\mathsf{2e} \otimes \mathsf{2e}$, if the hidden features are expanded up to $l_{\max} = 2$. However, as previously discussed, the $\mathsf{1o} \otimes \mathsf{1o}$ path is blocked in a symmetric environment due to complete cancellation in the $\mathsf{1o}$ channel under sum-aggregation. Meanwhile, the $\mathsf{2e} \otimes \mathsf{2e}$ path fails because of the fact that tensor products

between two equivalent entries cannot produce an antisymmetric output, as this violates the required symmetry structure. This explains the poor performance in predicting $\boldsymbol{Z}^*_{\mathrm{O}}$ in Al_2O_3 and SiO_2 when the "mix" module is employed, even with hidden features expanded up to $l_{\max} = 3$. Meanwhile, using the "tp" module clearly improves the prediction of $\boldsymbol{Z}^*_{\mathrm{O}}$ with $l_{\max} = 2$ setting, as the 1e channel can be generated from non-equivalent entries, such as $1e \otimes 2e$. However, the prediction of $\boldsymbol{Z}^*_{\mathrm{Al}}$ and $\boldsymbol{Z}^*_{\mathrm{Si}}$ remains unsatisfactory under the same setting, which suggests that the 2e channel in their local representations is itself constrained by the high site symmetry, and cannot adequately encode the full local charge response without proper inversion sensitivity. In other words, the $l_{\max} = 2$ representations fail to fully reflect the local symmetry, as sufficient inversion-sensitive information is still lacking. This limitation arises because higher-order odd-parity channels (e.g., $l \geq 3$) are excluded, while these channels are necessary to fully resolve the system's response under inversion. Hence, physically accurate predictions for all atomic sites, including $\boldsymbol{Z}^*_{\mathrm{Al}}$ and $\boldsymbol{Z}^*_{\mathrm{Si}}$, are only possible when both the "tp" module (enabling flexible, symmetry-adapted tensor products) and $l_{\max} = 3$ hidden features (providing non-trivial odd-parity channels like 3o) are employed.

Based on the above analysis, we conclude that resolving such symmetry-related issues requires two key ingredients: (1) the inclusion of non-trivial odd parity contributions in the graph representations and (2) the use of non-equivalent entries in tensor product for generating asymmetric features of the target tensor. In light of this, we propose a simple yet effective solution, termed the "mix2" module, for the above problems, which is particularly well-suited for $SE(3)$-equivariant models in predicting tensorial properties with pseudo components or of full $O(3)$ equivariance. As described in the main text, this "mix2" module introduces an additional equivariant linear transformation ($O(3)$-Linear) layer before the self-mix tensor product layer. This transformation generates distinct, non-equivalent entries in the self-mix tensor products, as illustrated in **Fig**.S2 (c). We emphasize that, although these transformations are termed $O(3)$-Linear,[2] they act exclusively on the polar tensor features produced by XPaiNN. As can be seen in **Fig**. S3, the "mix2" module, when used

with $l_{\max} = 3$ $SE(3)$-equivariant hidden features, achieves prediction accuracy and physical consistency on par with the "tp" module.

Using the proposed "tp" and "mix2" modules, we successfully trained and predicted tensorial properties for single systems with perturbed structures. The test performance results, summarized in **Table** S4, demonstrate that XPaiNN achieves superior accuracy compared to ETGNN, while the "tp" and "mix2" modules exhibit comparable predictive performance, confirming their effectiveness in capturing the required physical symmetries.

Table S4: Model test mean absolute errors for electric response properties of single systems with perturbed structures.

Property	Unit	System Formula	Data Size	ETGNN[a]	XPaiNN-tp	XPaiNN-mix2
$\boldsymbol{\epsilon}$	—	Al_2O_3	3190	5.54e-3	3.71e-3	3.72e-3
		HfO_2	10000	1.98e-3	1.22e-3	1.64e-3
		SiO_2	3992	2.86e-3	6.25e-4	7.75e-4
$\boldsymbol{Z^*}$	e	Al_2O_3	3190	1.18e-2	8.31e-3	8.93e-3
		HfO_2	10000	4.28e-3	7.83e-4	1.15e-3
		SiO_2	3992	1.94e-3	1.32e-3	1.45e-3
$\boldsymbol{e}$	C/m^2	AlN	3211	3.29e-4	7.33e-5	6.50e-5
		CdS	3000	4.69e-4	6.43e-5	7.48e-5
		ZnO	3000	5.59e-4	6.20e-5	7.17e-5

[a] Results are taken from Ref. 13.

We also note here that learning and predicting the dielectric tensor $\boldsymbol{\epsilon}$ does not encounter such parity-related issues, as shown in **Fig**. S3. The reason lies in the fact that $\boldsymbol{\epsilon}$ is a symmetric tensor, and its Irreps under $O(3)$ consist only of even-l channels with even parity ($1\times0e + 1\times2e$). These correspond to polar tensor components and can be captured within $SO(3)$-equivariant frameworks without requiring explicit inversion symmetry resolution. The same reasoning applies to learning and predicting the elastic tensor ($2\times0e + 2\times2e + 1\times4e$).

Additional results on crystal electric response

Table S5 further compares the performances of XPaiNN combined with various output modules in predicting $\boldsymbol{e}$ and $\boldsymbol{Z}^*$, both being partial symmetric or asymmetric tensors containing pseudo-tensor contributions. The evaluation is performed on the datasets used in the main text, i.e., the curated Jarvis-DFT DFPT dataset[16] and the BEC benchmark set.[19] As shown in **Table** S5, the original "mix" design achieves reasonably good performances in predicting $\boldsymbol{e}$, when hidden equivariant features are expanded up to $l_{\max} = 3$. However, it struggles significantly in predicting $\boldsymbol{Z}^*$, as noted in the main text and the reason, as explained in the previous section, is its inability to properly model antisymmetric contributions. Notably, the "mix2" module achieves performance on par with the "tp" module. The slightly lower errors observed for the "tp" module are likely due to its larger number of trainable parameters, rather than a fundamental advantage in symmetry expressiveness.

Table S5: Performances of XPaiNN combined with various output modules in predicting $O(3)$-equivariant tensorial properties.

Property	Datasets[a]	Metrics [Unit]	XPaiNN-mix	XPaiNN-mix2[b]	XPaiNN-tp
$\boldsymbol{e}$	Jarvis DFT	Fnorm ↓ [C/m²]	0.24	0.24	0.24
		EwT 25% ↑ [%]	58.7	59.9	**62.0**
		EwT 10% ↑ [%]	46.6	51.5	**53.4**
		EwT 5% ↑ [%]	44.0	46.6	**50.0**
$\boldsymbol{Z}^*$	perovskite	MAE [e]	0.0388	0.0120	**0.0116**
		RMSE [e]	0.0890	0.0200	**0.0195**
$\boldsymbol{Z}^*$	Li_3PO_4	MAE [e]	0.0327	0.0133	**0.0127**
		RMSE [e]	0.0497	0.0198	**0.0193**
$\boldsymbol{Z}^*$	ZrO_2	MAE [e]	0.0457	0.0199	**0.0185**
		RMSE [e]	0.0741	0.0430	**0.0420**

[a] Datasets are from Ref. 16 and Ref. 19. Best results are in **bold**.
[b] Results have been reported in the main text.

Table S6 reports the results of XPaiNN fitted to the original JARVIS-DFT DFPT dataset,[20] including the properties of $\boldsymbol{\epsilon}$, $\boldsymbol{e}$ and $\boldsymbol{Z}^*$. The results are compared with the regression model reported in the original work.[20] Both $\boldsymbol{\epsilon}$ and $\boldsymbol{e}$ were trained with the "mix" module, while $\boldsymbol{Z}^*$ was trained with the "tp" module. As noted in the main text, the raw

dataset contains a certain proportion of outliers as well as distorted structures. Therefore, we limited the total number of samples by data cleaning to ensure reliable and physically meaningful results; see **Experimental Section** for more information.

Table S6: Model performances in predicting electric response properties of the JARVIS-DFT DFPT dataset.

			XPaiNN		Regression Model [a]	
Property	Unit	Num. Crystal [b]	MAE	R^2	MAE	R^2
$\boldsymbol{\epsilon}$		4134	0.86	0.87	—	—
$\epsilon_{\max}$			2.92	0.63	4.91	< 0.2
$\boldsymbol{e}$	$\mathrm{C/m^2}$	4599	0.028	0.36	—	—
$e_{\max}$	$\mathrm{C/m^2}$		0.095	0.46	0.47	< 0.2
$\boldsymbol{Z}^*$	e	4807	0.14	0.83	—	—
$Z^*_{\max}$	e		0.20	0.83	0.60	0.76

[a] Results are taken from Ref. 20.
[b] Total number of samples after data cleaning for the target property.

Our present XPaiNN model shows fairly good accuracy for dielectric tensor and BEC, with the R square values (R^2) exceeding 0.80 for both. In comparison, the baseline regression model trained directly on the maximum component of each tensor label reports an R^2 of only 0.76 for $Z^*_{\max}$ and fails to resolve the correlation for $\epsilon_{\max}$. When the maximum component is extracted post-hoc from the full tensor predicted by XPaiNN, our model still outperforms the regression model in both MAE and R^2, where performances on $\boldsymbol{Z}^*$ and $Z^*_{\max}$ are particularly close.

However, our model seems not to improve over the regression model in piezoelectric tensor prediction, where a poor R^2 correlation between prediction and the ground truth is observed. In light of the above discussion, we attribute this limitation not to our approach itself, but primarily to insufficient or poorly qualified training data, which constrains the model's ability to learn the structure-property relationship governed by crystal symmetry. The piezoelectric tensor is inherently a sparse tensor in terms of its non-zero components. In systems possessing inversion symmetry, for example, the piezoelectric tensor must be zero.[21] Upon inspection, we found that approximately half of the samples in the raw piezoelectric dataset are labeled

with all-zero tensors, which limits the model's capacity to capture the exact piezoelectric behavior in other systems.

Fig. S4 shows the prediction results for all three properties on the test set of the JARVIS-DFT dataset in the current work, with the left column being the predicted tensor, while the right column corresponds to predictions for the maximum tensor component.

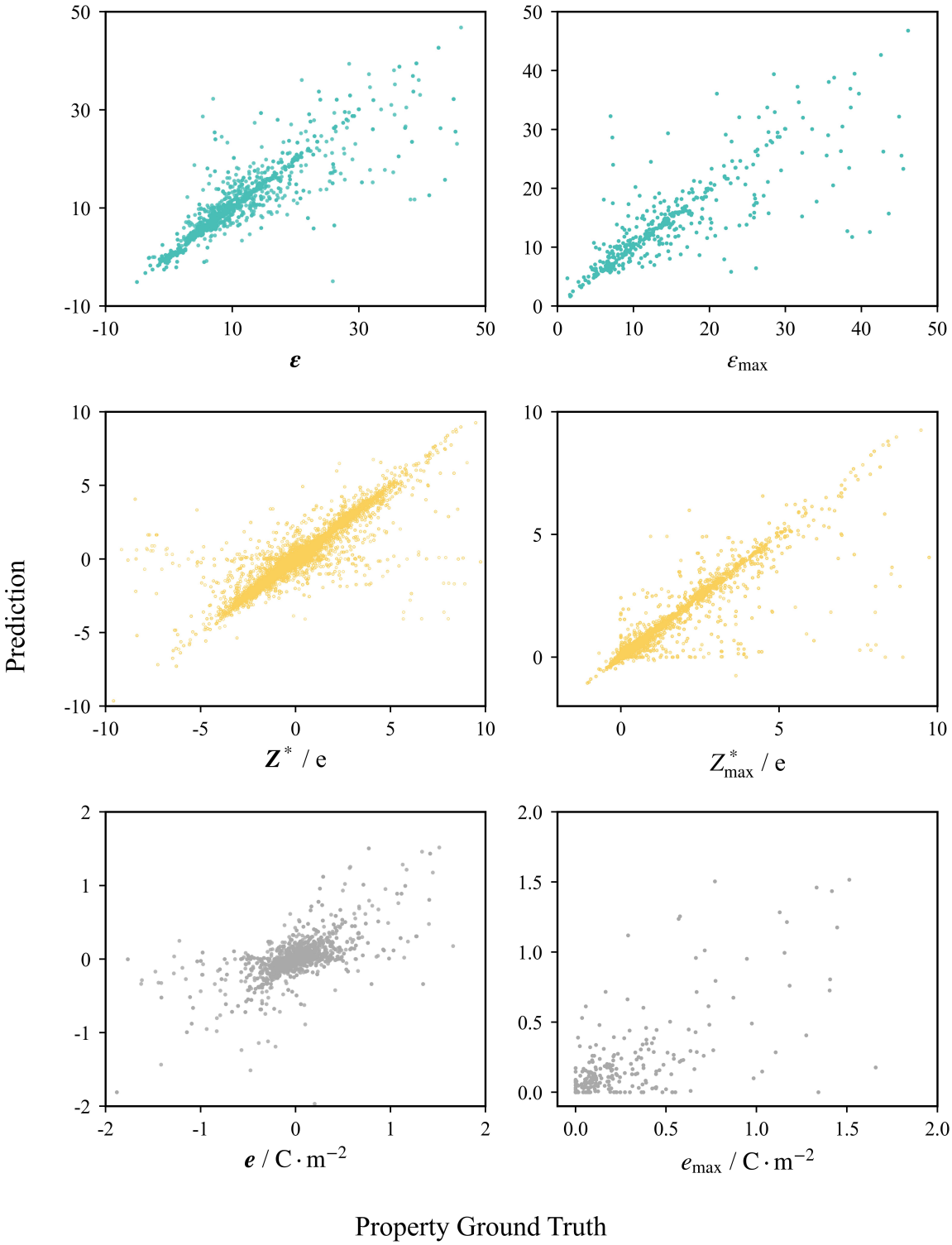


Figure S4: Plots of the predicted tensor properties and their maximum components in test set of the JARVIS-DFT DFPT dataset against the ground truth, with the left column being the predicted tensor and the right column the predicted maximum component.

Results of the MatTen dataset

Training and predicting the elastic tensor $\boldsymbol{C}$ are carried out in its fourth-rank Cartesian form, i.e., the model output is an 81-dimensional tensor of shape $(3, 3, 3, 3)$. The tensor is then transformed to a symmetric 6×6 matrix $\boldsymbol{c}$ in the Voigt notation c_{ij}, where $i, j \in \{1, 2, 3, 4, 5, 6\}$. The bulk modulus K, shear modulus G, Young's modulus E and Poisson's ratio v are then evaluated according to the Voigt-Reuss-Hill scheme (VRH).[9] The compliance tensor $\boldsymbol{s}$ in the Voigt notation is obtained as the inverse of $\boldsymbol{c}$

$$\boldsymbol{s} = \boldsymbol{c}^{-1} \tag{S12}$$

The Voigt and Reuss bulk/shear moduli are thus evaluated as

$$\begin{cases} K_{\mathrm{V}} = [(c_{11} + c_{22} + c_{33}) + 2(c_{12} + c_{23} + c_{31})] \ / \ 9 \\ K_{\mathrm{R}} = 1 \ / \ [(s_{11} + s_{22} + s_{33}) + 2(s_{12} + s_{23} + s_{31})] \\ G_{\mathrm{V}} = [(c_{11} + c_{22} + c_{33}) - (c_{12} + c_{23} + c_{31}) + 3(c_{44} + c_{55} + c_{66})] \ / \ 15 \\ G_{\mathrm{R}} = 15 \ / \ [4(s_{11} + s_{22} + s_{33}) - 4(s_{12} + s_{23} + s_{31}) + 3(s_{44} + s_{55} + s_{66})] \end{cases} \tag{S13}$$

K, G, E and v in this work are computed according to the Hill average scheme.

$$\begin{cases} K_{\mathrm{H}} = (K_{\mathrm{V}} + K_{\mathrm{R}}) \ / \ 2 \\ G_{\mathrm{H}} = (G_{\mathrm{V}} + G_{\mathrm{R}}) \ / \ 2 \\ E_{\mathrm{H}} = 9K_{\mathrm{H}}G_{\mathrm{H}} \ / \ (3K_{\mathrm{H}} + G_{\mathrm{H}}) \\ v_{\mathrm{H}} = (3K_{\mathrm{H}} - 2G_{\mathrm{H}}) \ / \ (6K_{\mathrm{H}} + 2G_{\mathrm{H}}) \end{cases} \tag{S14}$$

Table S7 compares the performance of the XPaiNN models to other results on the MatTen dataset. The results of the MatSca and AutoMatiner models are fitted to the isotropic elastic properties directly, while those of MatTen are derived from the predicted elastic tensor in the

same manner as XPaiNN.

Table S7: Model performances in predicting bulk, shear and Young's moduli in the MatTen test set.

	K		G		E	
Model	MAE[a]	MAE/MAD[a]	MAE	MAE/MAD	MAE	MAE/MAD
XPaiNN[b]	5.40	0.095	6.33	0.211	14.74	0.197
XPaiNNs[c]	5.21	0.092	6.24	0.208	14.55	0.194
	(1.60)	—	(1.73)	—	(3.94)	—
MatTen[d]	7.37	0.130	8.38	0.280	20.59	0.275
MatSca[d]	7.32	0.129	8.63	0.288	19.87	0.265
AutoMatiner[d]	9.84	0.174	9.27	0.309	22.10	0.295

[a] MAE for mean absolute error, unit in GPa. MAD stands for mean absolute deviation, calculated between prediction and the mean of reference, see Ref. 9.
[b] The best XPaiNN model test on the dataset, as is reported in the main text.
[c] Ensemble model of four, the averaged standard deviations over test samples are provided in the parentheses below.
[d] Results are taken from Ref. 9.

Table S8 shows the XPaiNN model performances trained with different settings of Spherical feature Irreps and Output hidden Irreps. Please refer to the **Experimental Section** for hyperparameter settings of the XPaiNN models. As can be seen, the model with $l_{\max} = 4$ achieves the best model performance. Meanwhile, the model with $l_{\max} = 3$ is also competitive, representing a notable progress as compared to that with $l_{\max} = 2$.

Table S8: XPaiNN model performances in predicting elastic properties in the MatTen test set with different settings of Irreps layout. Unit in GPa.

	K		G		E	
Model	MAE	RMSE[a]	MAE	RMSE	MAE	RMSE
XPaiNN ($l_{\max} = 2$)	5.96	11.44	7.20	12.07	16.86	28.49
XPaiNN ($l_{\max} = 3$)	5.53	11.02	6.43	11.40	14.92	26.67
XPaiNN ($l_{\max} = 4$)[b]	5.40	10.45	6.33	11.23	14.74	26.36

[a] Root mean squared error.
[b] Model hyperparameters used in this study and the corresponding performance results.

Table S9 and **Table** S10 report the success rates of the models in correctly predicting zero-valued elements and mutually dependent elements of the 6×6 stiffness matrix on the test set of the MatTen dataset. These predictions are subject to constraints imposed by crystal symmetries,[9] and the high success rates demonstrate the ability of our approach to construct tensors that satisfy intrinsic symmetry across various crystalline systems.

Table S9: Model prediction for symmetry-constrained zero-valued tensor elements in the stiffness matrix across various crystal systems. Success rate measured by error $< 10^{-4}$.

	XPaiNN-mix						
Crystal System	$l_{\max} = 2$	$l_{\max} = 3$	$l_{\max} = 4$[a]	XPaiNN-gated[a]	$SE(3)$-Test-mix[a]	$SE(3)$-Test-lin[a]	label[b]
Cubic	98.73%	98.71%	98.71%	98.10%	97.80%	97.70%	100.00%
Tetragonal	97.61%	97.56%	97.69%	94.10%	95.51%	93.59%	99.06%
Hexagonal	98.86%	98.80%	98.98%	94.00%	94.42%	94.66%	100.00%
Trigonal	93.33%	93.33%	93.19%	82.52%	87.11%	82.67%	96.44%
Orthorhombic	98.75%	98.81%	98.56%	96.37%	96.43%	95.68%	100.00%
Monoclinic	95.68%	95.68%	95.00%	94.55%	94.77%	94.55%	100.00%

[a] Models in **Table** S3.
[b] Success rate analysis of the ground truth.

Table S10: Model success rate in identifying symmetry-constrained mutually dependent elements of equality and anti-equality, measured by difference $< 10^{-3}$.

	XPaiNN-mix						
Crystal System	$l_{\max} = 2$	$l_{\max} = 3$	$l_{\max} = 4$[a]	XPaiNN-gated[a]	$SE(3)$-Test-mix[a]	$SE(3)$-Test-lin[a]	label[b]
Cubic	96.78%	96.83%	96.62%	96.49%	96.41%	96.01%	100.00%
Tetragonal	64.79%	64.79%	65.13%	64.79%	64.79%	64.96%	66.50%
Hexagonal	56.59%	57.79%	57.07%	57.79%	60.19%	56.35%	65.23%
Trigonal	72.67%	72.67%	72.67%	72.00%	72.00%	72.00%	74.33%

[a] Models in **Table** S3.
[b] Success rate analysis of the ground truth.

As evident from the results, models equipped with different output modules, i.e., the "mix", "lin" and "gated", show comparable performances in capturing the intrinsic symmetry of elastic tensors that adhere to crystal symmetry. Notably, XPaiNN combined with the "mix" module developed in this work achieves the best overall performance, regardless of the different settings of hidden equivariant feature Irreps. This advantage is particularly pronounced in trigonal systems, where the model excels at identifying zero-valued components. The success rate analysis of the ground truth suggests that minor structural distortions in the

input crystal geometry may prevent the model from generating a perfect symmetry-preserving graph representation. Such imperfections can propagate through the model, limiting its ability to fully recover exact symmetry relations, even when the model architecture is capable of representing them.

To conclude, the above results clearly support the arguments presented in the previous sections and the main text, demonstrating that our approach is capable of predicting tensors with intrinsic symmetry. This is particularly evident for properties within the $SO(3)$-equivariant framework.

Additional results on materials screening

Table S11 summarizes the XPaiNN model-predicted bulk and shear moduli on the recently reported ultraincompressible materials in the OQMD database, obtained using the Atomic Stiffness method.[22] As is clear, our results, averaged over four XPaiNN models, agree well with the previous work.[22] For most structures, the differences are within 10 GPa in method-predicted bulk modulus K, demonstrating the consistent and reliable predictive power of our method.

Table S11: Verification the XPaiNN-predicted bulk modulus K for the recently reported ultraincompressible materials. The XPaiNN-predicted shear modulus G is also included. Unit in GPa.

Material ID[a]	Formula	Atomic Stiffness[b]	DFT[b]	Liter.[b]	K^c	G^c
1230026	IrOs	376	416		367	202
1230666	ReOs	388	415		382	187
1435176	Re_3C	379	416		369	204
14924	C_3N_4	418	411	451	421	324
14925	C_3N_4	452	442	496	447	383
1625527	$MnIrOs_6$	385	418		364	214
1625528	$IrOs_6W$	385	410		374	216
1625529	$CrIrOs_6$	382	403		368	221
1625530	$ReIrOs_6$	394	422		387	225
1625532	$MnOs_6W$	377	401		354	200

[a] Identification in the OQMD database.

[b] Results of method-predicted, DFT calculated and literature reported bulk modulus, respectively, taken from Ref. 22.

[c] The XPaiNN-predicted bulk (K) and shear (G) moduli, derived and averaged over four models.

(**Table** S11 continued.)

Material ID[a]	Formula	Atomic Stiffness[b]	DFT[b]	Liter.[b]	K^c	G^c
1625533	$MnCrOs_6$	377	402		351	202
1625534	$MnReOs_6$	387	415		365	205
1625537	$ReOs_6W$	388	409		384	215
1625539	$CrReOs_6$	385	404		373	214
1625541	$MoReOs_6$	383	405		379	222
1625866	Re_6IrOs	373	401		372	189
21515	ReC	394	436	436–457	415	236
22207	Re_2C	384	428	389	374	213
22256	OsC	405	411	369	379	99
22307	ReN	410	419	453	415	210
22377	Re_2N	393	422	411	390	226
22378	Re_3N	386	417	400	395	218
301434	$ReOs_3$	400	416		393	218
301470	Re_3Os	382	401		382	195
301945	Os_3Pt	370	402		366	194
304348	$IrOs_3$	394	414		385	226
319188	Os_3Ru	386	406		380	244
319530	$IrOs_3$	394	419		387	227
323983	$TcOs_3$	383	403		384	246
325825	ReN	408	411		393	194

[a] Identification in the OQMD database.

[b] Results of method-predicted, DFT calculated and literature reported bulk modulus, respectively, taken from Ref. 22.

[c] The XPaiNN-predicted bulk (K) and shear (G) moduli, derived and averaged over four models.

(**Table** S11 continued.)

Material ID[a]	Formula	Atomic Stiffness[b]	DFT[b]	Liter.[b]	K^c	G^c
326001	IrOs	376	413		362	210
326779	ReC	391	407		384	192
327347	ReIr	360	402		356	219
327494	ReOs	387	409		374	184
336459	IrOs	377	403		366	216
337952	ReOs	388	410		379	192
344016	$IrOs_3$	390	413	421	377	222
345928	$MnOs_3$	378	401		333	201
347040	$ReOs_3$	395	417		384	209
8183	OsC	403	414	396	386	63

[a] Identification in the OQMD database.

[b] Results of method-predicted, DFT calculated and literature reported bulk modulus, respectively, taken from Ref. 22.

[c] The XPaiNN-predicted bulk (K) and shear (G) moduli, derived and averaged over four models.

Meanwhile, our method tends to provide lower values for the bulk modulus K, and the largest discrepancies are near 20 GPa for structures containing Mn. Besides, most of the

predicted values of the shear modulus G are above 180 GPa, except for two structures, whose K values are both above 350 GPa. In view of this, we set $K > 320$ GPa and $G > 180$ GPa as the criteria for our materials' screening study. Furthermore, an additional query of $K > 350$ GPa does not yield new candidates in the pool of 25,021 structures.

Fig. S5 is the zoomed plot of **Fig**. 5(b) in the main text. As can be seen, the concentrated area of scatter points indicate that most of the K values distribute between 1 and 200 GPa, while G values are between 1 and 125 GPa. Compound selection is achieved by sorting the compounds ($K \geq 320$ GPa; $G \geq 180$ GPa). 20 potentially different ultraincompressible, superhard materials are found, and those of boride and tungsten-containing compounds (BW) among these candidates are illustrated as red marks with star-shaped hollow, while the others are represented by yellow stars.

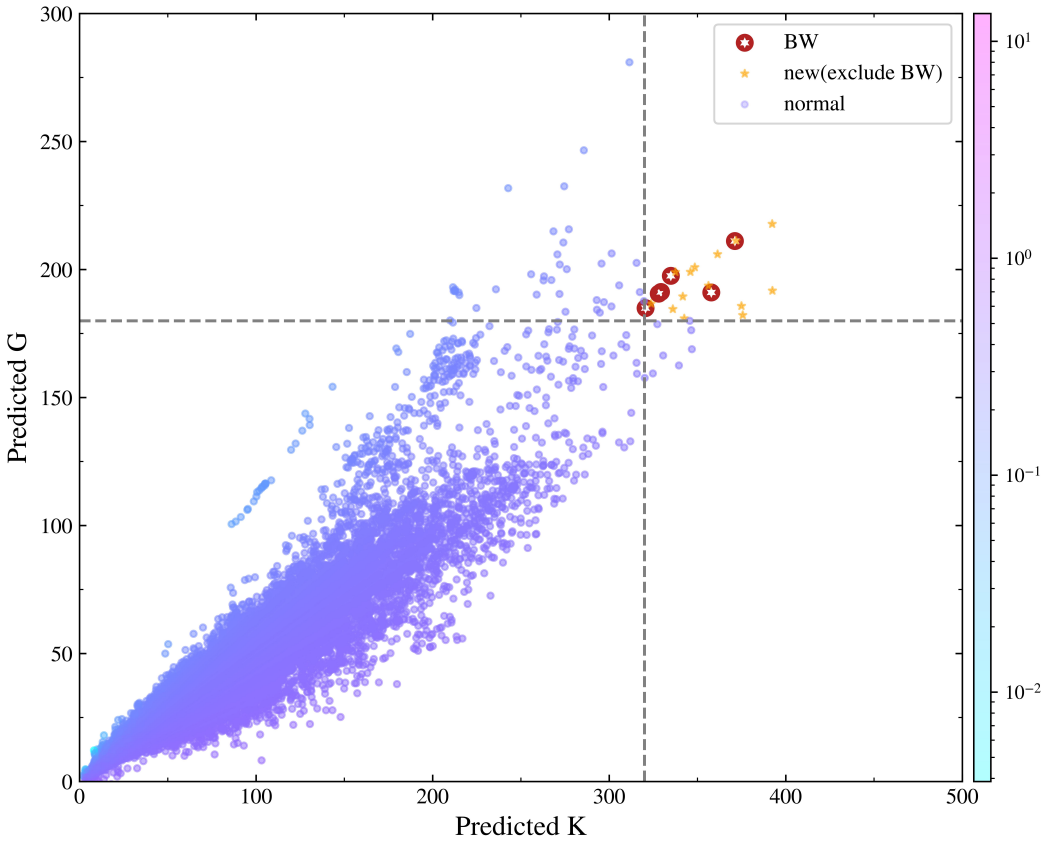


Figure S5: Plot of the XPaiNN-predicted G and K moduli of 25,021 structures, filtered from the Materials Project database.

Table S12 presents computational validations for the 6 predicted potentially ultraincompressible and superhard materials containing B and W, as identified by the present XPaiNN model. The first-principles calculations were performed by the Vienna Ab initio Simulation Package,[23] employing the generalized gradient approximation of the Perdew-Burke-Ernzerhof (PBE) functional.[24] An energy cutoff of 550 eV was applied, along with a tight self-consistent field (SCF) convergence criterion (EDIFF = 10^{-8} and EDIFFG = -0.001). K points for each system were generated with a k-point grid spacing value of 0.02 Å^{-1}. Other input settings were automatically generated using vaspkit.[25] Clearly, the XPaiNN predictions are in good agreement with the DFT results.

Table S12: Validation for material screening of ultraincompressible candidates containing B or W. Unit in GPa.

		XPaiNN		Calc.	
ID[a]	Formula	K_{VRH}	G_{VRH}	K_{VRH}	G_{VRH}
mp-1068296	$Fe(BW)_2$	328	191	334	199
mp-1070761	$Ni(BW)_2$	321	185	322	191
mp-1079348	B_3Os_2	335	198	344	196
mp-1104181	$Ni(BW)_3$	329	191	333	199
mp-1186374	Os_3W	371	211	372	225
mp-15671	Re_3B	358	191	365	231

[a] Identification in Materials Project database.

In **Fig**. S6, we consider the effect of two elements in unequal atomic fractions. Each data point on the graph satisfies the requirement that the number of the first element must be greater than or equal to the number of the second element in the pair combinations of elements. Here, the atomic numbers of the first element and the second element correspond to the row and column, respectively. Thus, this is a diagram designed with a rule of column-first. Specifically, in all possible combinations of elements as X_aY_b, it is ensured that a is greater than b. It can be seen from the asymmetry in the figure that the ratio of elements can also affect the bulk modulus, especially in the pair combinations of low atomic number elements with high atomic number elements. A combination of more low atomic number elements

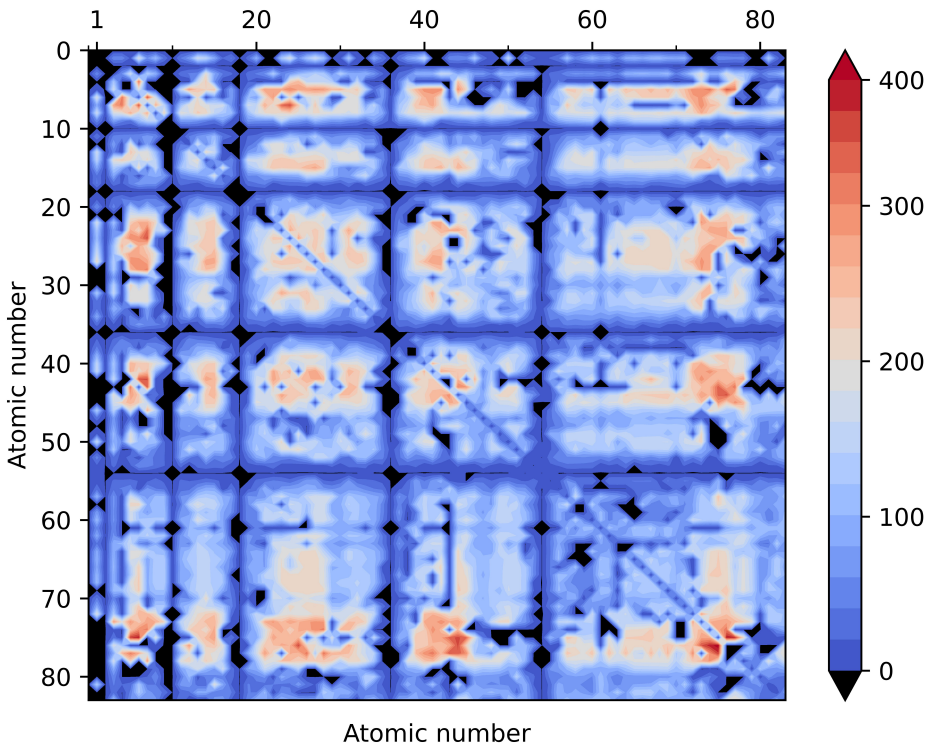


Figure S6: Correlation maps of element combinations for ultraincompressible materials, estimated based on the maximum K values ever observed in the pair combinations of elements in the ternary compounds.

with less high atomic number elements is recommended, which is more conducive to the construction of ultraincompressible materials.

Additional error statistics for QM9S and QM9NMR

Table S13 summarizes the detailed error statistics for the five tensor properties in the QM9S dataset for XPaiNN, while the results of DetaNet[17] are provided for comparison.

Table S13: Detailed comparison between XPaiNN and DetaNet on the QM9S tasks.

		DetaNet[a]			XPaiNN		
Property	Unit	MAE	RMSE	R^2	MAE	RMSE	R^2
$\boldsymbol{\mu}$	D	0.0089	0.1280	0.99994	0.0058	0.0098	0.99997
$\boldsymbol{\alpha}$	a_0^3	0.1367	0.2227	0.99996	0.0543	0.1001	0.99999
$\boldsymbol{\beta}$	a.u.	2.0373	2.8950	0.99398	1.4005	2.6736	0.99515
$\boldsymbol{D}$	D · Å	0.3834	0.6258	0.99945	0.2203	0.5988	0.99948
$\boldsymbol{O}$	D · Å^2	6.2442	13.116	0.97030	4.1617	15.0112	0.96191

[a] Results are taken from Ref. 17.

Table S14 summarizes models' performances for various quantities derived from the model-predicted $\boldsymbol{\mu}$, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. As is clear, results of the "mix" module are better than those of the "lin" and "gated" modules for the derived quantities, suggesting a better consistency in the predictions of different tensor components.

Table S14: Model performances for quantities derived from the predicted tensorial properties of QM9S.

Property	Metric [Unit]	XPaiNN-mix[a]	XPaiNN-mix[b]	XPaiNN-gated[b]	$SE(3)$-Test-mix[b]	$SE(3)$-Test-lin[b]
μ	MAE [Debye]	0.0065	0.0057	0.0065	0.0132	0.0185
	RMSE [Debye]	0.0113	0.0096	0.0114	0.0194	0.0268
α_{iso}	MAE [a_0^3]	0.0519	0.0567	0.0564	0.1362	0.1743
	RMSE [a_0^3]	0.0919	0.0932	0.0947	0.2303	0.2896
α_{aniso}	MAE [a_0^3]	0.0886	0.0834	0.0874	0.2374	0.2800
	RMSE [a_0^3]	0.1800	0.1620	0.1690	0.4224	0.4979
β_{norm}	MAE [a.u.]	4.9320	4.4557	4.9351	8.3130	11.4318
	RMSE [a.u.]	8.9794	8.3884	9.1006	17.4475	26.4236
$\beta_{\perp z}$	MAE [a.u.]	2.0985	1.8956	2.0972	3.4344	4.5393
	RMSE [a.u.]	3.6465	3.2160	3.5294	6.6383	9.3634
$\beta_{\parallel z}$	MAE [a.u.]	0.6995	0.6319	0.6991	1.1448	1.5131
	RMSE [a.u.]	1.2155	1.0720	1.1765	2.2128	3.1211

[a] The MAE results of the derived α and β properties are reported in **Fig**. 2 in the main text.
[b] Models in **Table** S2.

The derived quantities are calculated according to

$$\mu = \sqrt{\mu_x^2 + \mu_y^2 + \mu_z^2} \tag{S15}$$

$$\alpha_{\text{iso}} = 1 \ / \ 3 \ (\alpha_{xx} + \alpha_{yy} + \alpha_{zz}) \tag{S16}$$

$$\alpha_{\text{aniso}} = \sqrt{(\alpha_{xx} - \alpha_{yy})^2 + (\alpha_{yy} - \alpha_{zz})^2 + (\alpha_{zz} - \alpha_{xx})^2} \tag{S17}$$

$$\beta_{\text{norm}} = \sqrt{\beta_x^2 + \beta_y^2 + \beta_z^2} \tag{S18}$$

$$\beta_{\parallel z} = 3 \ / \ 5 \ \beta_z \tag{S19}$$

$$\beta_{\perp z} = 1 \ / \ 5 \ \sum_j (2\beta_{zjj} - 3\beta_{jzj} + 2\beta_{jjz}) \tag{S20}$$

$$\beta_i = 1 \ / \ 3 \ \sum_j (\beta_{ijj} + \beta_{jij} + \beta_{jji}) \quad i, j \in \{x, y, z\} \tag{S21}$$

Table S15 reports error statistics for each nucleus as well as all sites in the test set of QM9NMR in the current study. Results of DetaNet (using four element-wise models) and NMRNet (joint model) are employed for comparison.

Table S15: Error statistics on the shielding constants in the QM9NMR dataset. Unit in ppm.

	DetaNet[a]			XPaiNN			NMRNet[a]		
Nuclei	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE	RMSE	R^2
^{1}H	0.0543	0.0778	0.9985	0.0945	0.1594	0.9936	0.0478	0.0708	0.9987
^{13}C	0.5203	0.8822	0.9997	0.4362	0.7076	0.9998	0.3348	0.5611	0.9999
^{15}N	0.9866	1.7183	0.9998	0.8775	1.7509	0.9998	0.8653	1.6396	0.9998
^{17}O	1.6982	2.7342	0.9998	1.3567	2.8797	0.9998	1.2780	2.3063	0.9999
^{19}F	—	—	—	0.8702	1.2708	0.9985	0.7969	1.1339	0.9988
all	0.4938	1.1322	0.9998	0.3583	1.0053	0.9999	—	—	—

[a] Results are taken from Ref. 17 and Ref. 26.

Training efficiency

Different tensorial properties have distinct layouts of output Irreps due to their tensor order and fundamental symmetry. Hence, models trained for different properties have varying numbers of trainable parameters, even when using the same output approach and within the same model architecture. These differences are further amplified by the choice of output modules described above. Meanwhile, we shall point out that it is the model architecture that affects the training efficiency the most, particularly how the message passing blocks are implemented. To illustrate this, we compare the training time per epoch across several models, including (1) XPaiNN equipped with either the "mix" or "mix2" (for $\boldsymbol{e}$ and $\boldsymbol{Z}^*$) modules; (2) the standard $E(3)/SE(3)$-equivariant GNN model (based on the Test Model architecture) combined with the "lin" output module; (3) the XPaiNN-gated combination for the polar tensor properties. For fair comparison, we kept the hidden equivariant feature dimensions identical for all models within the same task, with two exceptions for $\boldsymbol{e}$ and $\boldsymbol{Z}^*$. For $\boldsymbol{e}$, the hidden feature dimension of the $E(3)$-Test Model was set to 64, half that of XPaiNN (128), to ensure the same total number of feature channels. For $\boldsymbol{Z}^*$, the hidden equivariant features of the Test Model were expanded up to $l_{\max} = 2$ to align with the benchmark BM model.[19] For more details about the hyperparameters, see the **Experimental Section**. All trainings were performed on 2 NVIDIA RTX3090 GPUs.

As shown in **Table** S16, the number of training parameters for each model is property-specific, and generally increases with the tensor order. In fact, the number of parameters in the "lin" and "gated" modules scales linearly with the total number of channels in the tensor's Irreps, or equivalently, the number of independent tensor components. In the case of the "mix" module, the number of parameters grows faster, which scales quadratically with the number of channels due to the self-mix tensor product operation.

In terms of training efficiency, the XPaiNN architecture clearly shows a significant advantage over the Test Model, an architecture that follows a conventional $SE(3)/E(3)$-equivariant GNN. For molecular properties, even though the number of trainable parameters

Table S16: Trainable parameters and training efficiency for different model architectures and output modules.

				XPaiNN-mix[a]		Test-lin[a]		XPaiNN-gated[a]	
Property	Order	#Samples[b]	Batch size	#Param.	Time/Epoch (s)	#Param.	Time/Epoch (s)	#Param.	Time/Epoch (s)
$\boldsymbol{\mu}$	1	110,000	100	916,756	115.2	517,424	114.4	896,213	105.4
$\boldsymbol{\alpha}$	2	110,000	100	927,156	123.9	517,456	118,8	896,342	107.0
$\boldsymbol{\beta}$	3	110,000	100	929,236	119.8	517,456	120.5	896,342	103.3
$\boldsymbol{e}$	3	3,907	32	1,795,988	18.7	1,408,912	74.1	—	—
$\boldsymbol{Z^*}$	2	23,456[c]	32	941,588	116.8	459,120	311.2	—	—
$\boldsymbol{C}$	4	8,230	32	2,194,612	41.4	1,940,016	96.4	2,093,977	39.6

[a] For the $O(3)$-equivariant properties, $\boldsymbol{e}$ and $\boldsymbol{Z^*}$, XPaiNN-mix corresponds to the "mix2" variant. The Test Model is $E(3)$-equivariant. XPaiNN combined with the "gated" module is not applicable to $O(3)$-equivariant properties.
[b] Number of molecules and crystals in the training set.
[c] There are 980 samples from the perovskite dataset, 14,393 from the Li_3PO_4 dataset and 8,083 from the ZrO_2 dataset.

of XPaiNN is approximately double compared to that of the Test Model, the training time per epoch is comparable or even shorter. The reason is that expensive tensor product convolutions during message passing are avoided in the XPaiNN architecture. Notably, the efficiency gain is more pronounced in modeling properties of crystalline systems, where unit cells typically involve larger and more densely connected graphs due to the periodic boundary condition. This advantage is even more evident for $O(3)$-equivariant properties, where tensor product convolutions become even more expensive in $E(3)$ models. This significantly increased computational cost is largely due to the $O(L^6)$ bottleneck,[11] where L refers to the maximum angular momentum of the hidden equivariant features.

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Experimental Section

Defining an XPaiNN model requires specifying several architectural hyperparameters. For the hidden layers, these include setting dimensions for $\boldsymbol{x}$ (Scalar feature dimension) and Irreps layout (Spherical feature Irreps) for $\boldsymbol{\chi}$ in model hidden layers. For the Cartesian output layer, additional settings are required, which include Output hidden dimension and Output hidden Irreps for transformations within the output module. The final output of the tensor property is determined by two key physical parameters: tensor Order and fundamental Symmetry, as summarized in **Table** 6 in the main text. Other architectural hyperparameters include: Number of radius basis for the expansion of the radial basis function, Cutoff radius for the adjacency of the graph, Number of blocks for the depth of NN, Layer normalization to specify whether to apply normalization in message passing blocks and Reduce operation for ways of pooling acting on the node output. For all models trained and tested in this work, we use the 0-order spherical Bessel function of the first kind enveloped by cosine cutoff functions as the radial basis functions. Other settings are related to model training.

Similarly, the Test Model implemented in this work requires settings of Spherical feature Irreps to distinguish its group equivariance ($E(3)$ or $SE(3)$). For instance, 64×0e + 64×1o + 64×2e stands for the setting of an $SE(3)$-equivariant model with hidden equivariant feature expanded up to $l_{\max} = 2$, while 64×0e + 64×0o + 64×1e + 64×1o + 64×2e + 64×2o represents the corresponding $E(3)$-equivariant model. All other architectural settings for the Test Model are identical to those used for XPaiNN.

To switch between Output module in our implementation, we use the following keyword flags: "cart_tensor" stands for the original "mix" module depicted in **Fig**.S1 (c), as well as **Fig**.1 (c) of the main text, "cart_tensor_lin" and "cart_tensor_gate" represent the "lin" and "gated" modules in **Fig**.S1 (a) and (b), respectively. The "mix2" module in **Fig**.S2 (c) is employed by calling "cart_tensor_mix", while the "tp" module in **Fig**.S2 (a) is activated by "cart_ten_tp". Moreover, for the layout of the hidden $O(3)$-equivariant feature in the "tp" module, which is expanded according to **Eq**. (S3), both Maximum L and Output Irreps

dimension should be explicitly defined.

QM9S

The five tasks in QM9S share the same random partition of the dataset into training, validation and testing subsets, which contain approximately 110 k, 10 k and 10 k (the remainder) samples, respectively. All five models use the same set of hyperparameters, except for Order and Symmetry, which are property-specific and lead to some minor differences in the total number of model trainable parameters.

Table S17: Model hyperparameters for training the QM9S tasks.

hyperparameters	Setting
Scalar feature dimension	128
Spherical feature Irreps	128×0e+64×1o+32×2e
Output module	cart_tensor
Output hidden dimension	64
Output hidden Irreps	32×0e+32×1o+32×2e
Number of radius basis	20
Cutoff radius (Å)	5.0
Layer normalization	False
Number of blocks	3
Reduce operation	Sum
Batch size	100
Training epochs	500
Learning rate warmup	Linear warmup for 25 epochs
Maximum learning rate	5×10^{-4}
Minimum learning rate	1×10^{-7}
Learning rate scheduler	Cosine annealing with Tmax = 500
Loss function	Smooth L1
Optimizer	Adam
Exponential moving average decay	0.995

For the $SE(3)$-Test Model, the following hyperparameters are utilized. The Scalar feature dimension is set for the linear transformation in embedding layer, which is in agreement with number of scalar channels in Spherical feature Irreps.

Table S18: Hyperparameters of the Test Model for training the QM9S tasks.

hyperparameters	Setting
Scalar feature dimension	64
Spherical feature Irreps	64×0e+64×1o+64×2e+64×3o
Output hidden Irreps	32×0e+32×1o+32×2e+32×3o
Number of radius basis	16

For comparison studies presented in **Table** S2, the Spherical feature Irreps are set to 64×0e+64×1o+64×2e+64×3o, and Output hidden Irreps to 32×0e+32×1o+32×2e+32×3o for both XPaiNN and the $SE(3)$-Test model. These models are equipped with different output modules, by switching the Output module keyword among "cart_tensor", "cart_tensor_lin" and "cart_tensor_gate". All other hyperparameters are the same as those listed in **Table** S17 and **Table** S18.

QM9NMR and nmrshiftdb2

We adopt an 80/10/10 ratio for the random split of the QM9NMR dataset into training, validation and test subsets. The model is directly fitted to the atom-wise chemical shielding constants by taking averaged trace of the atomic order-2 tensor output during both training and testing.

$$\sigma_A = \frac{1}{3}\mathrm{Tr}\left(\boldsymbol{T}_A\right) \tag{S22}$$

Table S19: Model hyperparameters for training QM9NMR.

hyperparameters	Setting
Scalar feature dimension	128
Spherical feature Irreps	128×0e+64×1o+32×2e
Output module	cart_tensor
Output hidden dimension	64
Output hidden Irreps	32×0e+32×1o+32×2e
Number of radius basis	20
Cutoff radius (Å)	5.0
Layer normalization	False
Number of blocks	3
Reduce operation	None
Batch size	100
Training epochs	500
Learning rate warmup	Linear warmup for 25 epochs
Maximum learning rate	5×10^{-4}
Minimum learning rate	1×10^{-7}
Learning rate scheduler	Cosine annealing with Tmax = 500
Loss function	Smooth L1
Optimizer	Adam
Exponential moving average decay	0.995

For the task of predicting chemical shifts in nmrshiftdb2-2024, we use a Cutoff radius of 4.0 Å and apply Layer normalization for both direct training and fine-tuning. This means that a separate model is first trained using these settings on QM9NMR, which serves as the pretrained model. For the fine-tuning approach proposed in this work, we perform linear regression on the training set of nmrshiftdb2-2024, using the model predicted chemical

shielding constants, according to the relationship

$$\delta_A = \sigma^{\text{ref}} - 1.0 \times \sigma_A \tag{S23}$$

The resultant fitting parameters (the slope and the intercept) are then adopted as model hyperparameters for the chemical shift prediction task, with the Output module set to "chemical_shifts". **Fig**. S7 shows the linear regression results of ^{1}H and ^{13}C, demonstrating the feasibility of our fine-tuning approach. For ^{13}C, a clear linear relationship is observed. The results for ^{1}H are somewhat degraded due to outliers. Notably, the fitted intercepts closely resemble the reference shielding values commonly associated with TMS (tetramethylsilane) for these nuclei.

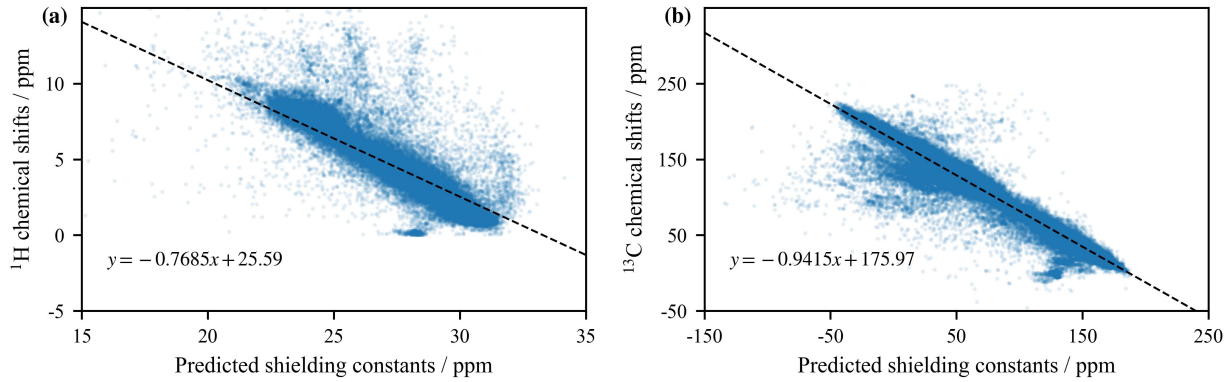


Figure S7: **Linear regression results** of **(a)** ^{1}H and **(b)** ^{13}C chemical shifts with respect to the pretrained model-predicted shielding constants.

Single systems with perturbed structures

Here we specify the hyperparameters for XPaiNN, when combined with the "tp" module. This module is activated by setting the Output module keyword along with Maximum L and Output Irreps dimension. These parameters are used to determine the hidden Irreps layout generated by the parity-expansion layer, as illustrated in **Fig**. S2. For the "mix" and "mix2" modules, these settings are similar: Spherical feature Irreps are kept the same as the "tp" module, and the Output hidden Irreps are set to 32×0e+32×1o+32×2e+32×3o. As with ETGNN, a 60/20/20 random split is used to form training, validation and testing subsets for each system.

Table S20: Model hyperparameters for training ETGNN study cases.

hyperparameters	Setting
Scalar feature dimension	128
Spherical feature Irreps	64×0e+64×1o+64×2e+64×3o
Output module	cart_tensor_tp
Output hidden dimension	64
Output Irreps dimension	32
Maximum L	3
Number of radius basis	20
Cutoff radius (Å)	6.0
Layer normalization	False
Number of blocks	3
Reduce operation	Sum, None for $\boldsymbol{Z}^*$
Batch size	32
Training epochs	2500
Learning rate warmup	Linear warmup for 25 epochs
Maximum learning rate	5×10^{-4}
Minimum learning rate	1×10^{-8}
Learning rate scheduler	Cosine annealing with Tmax = 500
Loss function	Smooth L1
Optimizer	Adam
Exponential moving average decay	0.995

JARVIS-DFT DFPT

Data cleaning. For the raw data, we first filter out crystals that contain noble gas elements (Group 18) and the actinide series. Element types of Tc, Cs, Eu and Gd are also excluded due to their rare occurrence. Moreover, value thresholds on the maximum absolute component $|\boldsymbol{T}|_{\max}$ are applied for filtering the tensor properties. The thresholds are 50.0 and 2.0 C/m^2 for dielectric tensor $\boldsymbol{\epsilon}$ and piezoelectric tensor $\boldsymbol{e}$, respectively. For BEC, crystals with any site that possesses $|\boldsymbol{Z}^*|_{\max}$ values above 15.0 e are excluded. This leads to datasets of 4134, 4599, 4807 systems for $\boldsymbol{\epsilon}$, $\boldsymbol{Z}^*$ and $\boldsymbol{e}$, respectively.

Training. For each dataset, the training, validation and testing subsets are formed using an 80/10/10 random split. The loss function employed is a combination of MAE (L1) and RMSE metrics.

$$\mathcal{L}\left(y, \hat{y}\right) = \frac{1}{N}\sum_i |y_i - \hat{y}_i| + \sqrt{\frac{\sum_i |y_i - \hat{y}_i|^2}{N}} \tag{S24}$$

where N denotes the number of total tensor components in a batch. Model hyperparameters are listed below, which are related to the results in **Table** S6. Note that we apply layer normalization and mean pooling readout for training $\boldsymbol{\epsilon}$, and we use the "tp" output module in predicting $\boldsymbol{Z}^*$. Other training settings are kept the same as described above.

Table S21: Model hyperparameters for training dielectric tensor in the raw JARVIS-DFT dataset.

hyperparameters	Setting
Scalar feature dimension	128
Spherical feature Irreps	128×0e+128×1o+128×2e+128×3o
Output module	cart_tensor
Output hidden dimension	64
Output hidden Irreps	32×0e+32×1o+32×2e+32×3o
Number of radius basis	20
Cutoff radius (Å)	6.0
Layer normalization	True
Number of blocks	3
Reduce operation	Mean
Batch size	32
Training epochs	2500
Learning rate warmup	Linear warmup for 25 epochs
Maximum learning rate	5×10^{-4}
Minimum learning rate	1×10^{-8}
Learning rate scheduler	Cosine annealing with Tmax = 500
Loss function	MAE + RMSE
Optimizer	Adam
Exponential moving average decay	0.995

Table S22: Model hyperparameters for training Born effective charge in the raw JARVIS-DFT dataset.

hyperparameters	Setting
Scalar feature dimension	128
Spherical feature Irreps	128×0e+128×1o+128×2e+128×3o
Output module	cart_tesnor_tp
Output hidden dimension	64
Output Irreps dimension	32
Maximum L	3
Number of radius basis	20
Cutoff radius (Å)	6.0
Layer normalization	False
Number of blocks	3
Reduce operation	None

Table S23: Model hyperparameters for training piezoelectric tensor in the JARVIS-DFT dataset.

hyperparameters	Setting
Scalar feature dimension	128
Spherical feature Irreps	128×0e+128×1o+128×2e+128×3o
Output module	cart_tensor
Output hidden dimension	64
Output hidden Irreps	32×0e+32×1o+32×2e+32×3o
Number of radius basis	20
Cutoff radius (Å)	6.0
Layer normalization	False
Number of blocks	3
Reduce operation	Sum

The curated Jarvis-DFT DFPT dataset used by GMTNet. We perform data cleaning on the curated dataset provided by GMTNet using the same criteria for element types. For numerical thresholds, a cutoff of 50.0 is applied to both dielectric and piezoelectric tensors (in C/m^2). The filtered dataset is then randomly split into training, validation and test sets using an 80/10/10 ratio. Here, the "mix2" output module is employed and the results are reported in the main text. Training is conducted with a maximum of 1,500 epochs, employing a cosine annealing learning rate scheduler (Tmax=300) and a linear warm-up over the first 15 epochs. For the $SE(3)$-Test Model, the Spherical feature Irreps and the Output hidden Irreps are kept the same as those used in XPaiNN. On the other hand, the $E(3)$-Test Model includes both even and odd parity channels in each l-order representations, while the feature dimensions are reduced from 128 to 64 per Irrep to ensure that the total number of trainable parameters remains comparable across all models.

Table S24: Model hyperparameters for training dielectric and piezoelectric tensor in the curated JARVIS-DFT dataset.

hyperparameters	Setting
Scalar feature dimension	128
Spherical feature Irreps	128×0e+128×1o+128×2e+128×3o
Output module	cart_tensor_mix
Output hidden dimension	64
Output hidden Irreps	32×0e+32×1o+32×2e+32×3o
Number of radius basis	20
Cutoff radius (Å)	6.0
Layer normalization	True
Number of blocks	3
Reduce operation	Mean
Batch size	32
Training epochs	1500
Learning rate warmup	Linear warmup for 15 epochs
Maximum learning rate	5×10^{-4}
Minimum learning rate	1×10^{-8}
Learning rate scheduler	Cosine annealing with Tmax = 300
Loss function	MAE + RMSE
Optimizer	Adam
Exponential moving average decay	0.995

Table S25: Comparison of dataset statistics between those employed in GMTNet and in the current work.

Dataset	# Samples	Fnorm Mean	Fnorm STD	Unit
Dielectric	4713	14.7	18.2	—
current	4508	12.6	11.0	—
Piezoelectric	4998	0.43	3.09	C/m^2
current	4884	0.35	1.39	C/m^2

Calculations of Frobenius norm for piezoelectric tensor are carried out in the 3×6 matrix form. In **Table** S5, predictions on piezoelectric tensor with a near-zero Fnorm value (as per the ground truth) are counted as high-quality (EwT) (in each case), if the norm distance between the prediction and the target is below the threshold of 10^{-4}.

The BEC benchmark dataset. The model hyperparameters are the same as those in **Table** S24, except for the Spherical feature Irreps, whose dimensions are reduced to 64 for each Irrep channel. For the Test Models, the $SE(3)$-mix2 model follows the setting of XPaiNN with 16 radius basis, while the $E(3)$-lin model sets 64×0e+64×1e+64×1o+64×2e for Spherical feature Irreps and 32×0e+32×1e+32×1o+32×2e for Output hidden Irreps, respectively. Following the original benchmark protocol, models are trained jointly on all three datasets with an 80/10/10 ratio of data splitting. The maximum number of training epochs is set to 300, using a cosine annealing learning rate scheduler (Tmax=300) with a linear warm-up over the first 10 epochs.

MatTen

In order to train the XPaiNN models on the whole dataset, we replace the node embedding from the default atom-wise scalar embeddings to one-hot embedding, as the default setting of XPaiNN does not support elements from the actinide series, such as Th and U. Dataset partitioning follows the same split scheme used in the original study, as provided by the authors. Model hyperparameters are largely identical to that of training the dielectric tensor in JARVIS-DFT, except for the Irreps layout. For results reported in **Table** S8, the model with $l_{\max} = 2$ refers to a setting of 128×0e+128×1o+128×2e for Spherical feature Irreps and 32×0e+32×1o+32×2e for Output hidden Irreps, respectively. A similar configuration applies for $l_{\max} = 3$, extended to include higher-order Irreps accordingly.

Table S26: Model hyperparameters for training elastic tensor in the MatTen dataset.

hyperparameters	Setting
Scalar feature dimension	128
Spherical feature Irreps	128×0e+128×1o+128×2e+128×3o+128×4e
Output module	cart_tensor
Output hidden dimension	64
Output hidden Irreps	32×0e+32×1o+32×2e+32×3o+32×4e
Number of radius basis	20
Cutoff radius (Å)	6.0
Layer normalization	True
Number of blocks	3
Reduce operation	Mean
Batch size	32
Training epochs	2500
Learning rate warmup	Linear warmup for 25 epochs
Maximum learning rate	5×10^{-4}
Minimum learning rate	1×10^{-8}
Learning rate scheduler	Cosine annealing with Tmax = 500
Loss function	MAE + RMSE
Optimizer	Adam
Exponential moving average decay	0.995

For the materials screening study, we query the Materials Project database for crystal

structures that do not have elasticity labels, consisting of elements whose atomic numbers are below 84 (Po) and excluding element types of rare gas, Cs and Pm. A stability criterion of $\Delta E_{\text{hull}} < 10^{-6}$ eV/atom is applied. The query is performed using the MatMiner package, resulting in a dataset of 25,021 structures. Upon further inspection, approximately 400 chemical formulas in the screening dataset are found to overlap with those in the training and validation sets. (This does not necessarily indicate exact structural duplicates, as these entries may correspond to different crystal structures (e.g., different space groups or polymorphs). However, a detailed structural comparison is not feasible within the current scope.)